Instrument I1 rerun with a calibrated A/A gate

This report follows up ../weakclass-20261104, where I1 (the character-sum maximum of the factor base's log set) came out **INVALID**. That run's preregistered A/A gate demanded that even and odd null means agree within 5% on every curve, which was too strict for 20 + 20 draws. Protocol protocol.json, WEAKCLASS-I1-20261104, was frozen before data (sha256 dd55966290eb..., commit ddf4a12a).

Changes from the first run

- **Calibrated A/A gate.** Each curve gets 200 null draws in two streams of 100, compared with a two-sample KS test. The gate passes iff the number of curves with $p < 0.05$ is within the 99th percentile of Binomial(#curves, 0.05). The KS routine was checked on synthetic data before the freeze: 6.0% false rejections, and 87% power against a 0.5 SD shift.
- **Fresh curves.** Picks are seeded by this protocol's hash. Only 4 of the 54 weak or paired curves coincide with the first run.
- **Out-of-sample overdispersion test.** 30 extra ordinary prime-order curves per size (90 in total) test the first run's descriptive observation that ordinary curves sat in the null's tails too often.
- The curve picking, factor bases, statistic and null law are all unchanged (weakclass.py is a byte copy of the frozen first-run script).

Results (frozen rule)

- **Controls valid.** 7 curves rejected A/A at $p < 0.05$, against an allowance of 14. The positive control (planted log-AP) has $u \geq 0.95$ on 100% of curves.
- **Weak classes: all BLIND.** The mean u is 0.46 for anomalous, 0.44 for small embedding degree and 0.28 for $j = 0$ with a ζ-stable factor base. No class is consistently beyond the 5–95% band at any size. The $j = 0$ class sits somewhat low (two curves at $u \leq 0.015$ at $L = 12$, one at 0.03 at $L = 14$), but not consistently. The first run's "$j = 0$ extreme low at two sizes" did not replicate on fresh curves.
- **Overdispersion: CALIBRATED.** 11 of 90 fresh ordinary curves are beyond 5%/95% (2 low, 9 high), against 9 expected, with a binomial tail of $p = 0.29$. The first run's apparent overdispersion (about 10 of 27 curves) came from its coarse 40-draw percentiles and chance.

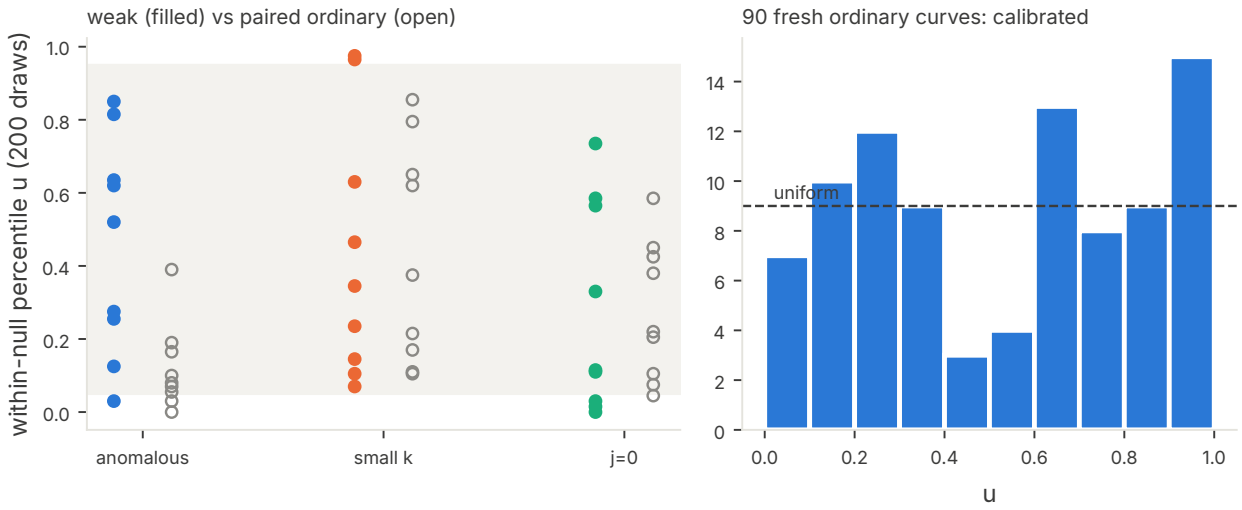

Figure 1: I1 rerun

Consequence for the weak-class table

With I1 now valid, **all three instruments (I1 character sum, I2 m = 2 sumset yield, I3 QTT) are BLIND** to anomalous, small-embedding-degree and $j = 0$ curves at $p \leq 2^{14}$. A NULL from any of them excludes only structure of that statistic's own type. It says nothing about whether a curve's DLP is easy. The I1 null of negation-closed random sets is calibrated for small-x log sets at $n \leq 2^{14}$.

Deviations

- No S3 checkpoint (no upload token). With the owner's approval, the pushed branch is the resume record.
- **Storage.** The full rows (585 KB) exceed the 64 KiB in-Git limit even under xz, and this session cannot write to the artifact store. rows-r3.jsonl.xz keeps every field, with the null draws rounded to 3 decimals. analyze_i1.py on it gives a verdict.json identical to the full-precision one. The full file's hash is in data.sha256, and a rerun from the frozen seeds reproduces it exactly.

Reproduce

```
python3 weakclass_i1.py --protocol-sha dd55966290eb11bd7a516640e3438e2f05c7936dd0a07c785f350c720e0b325c --out rows.jsonl
python3 analyze_i1.py rows.jsonl verdict.json      # or: xz -dc rows-r3.jsonl.xz > rows-r3.jsonl; same verdict
python3 figures.py; ./build_pdf.sh
```